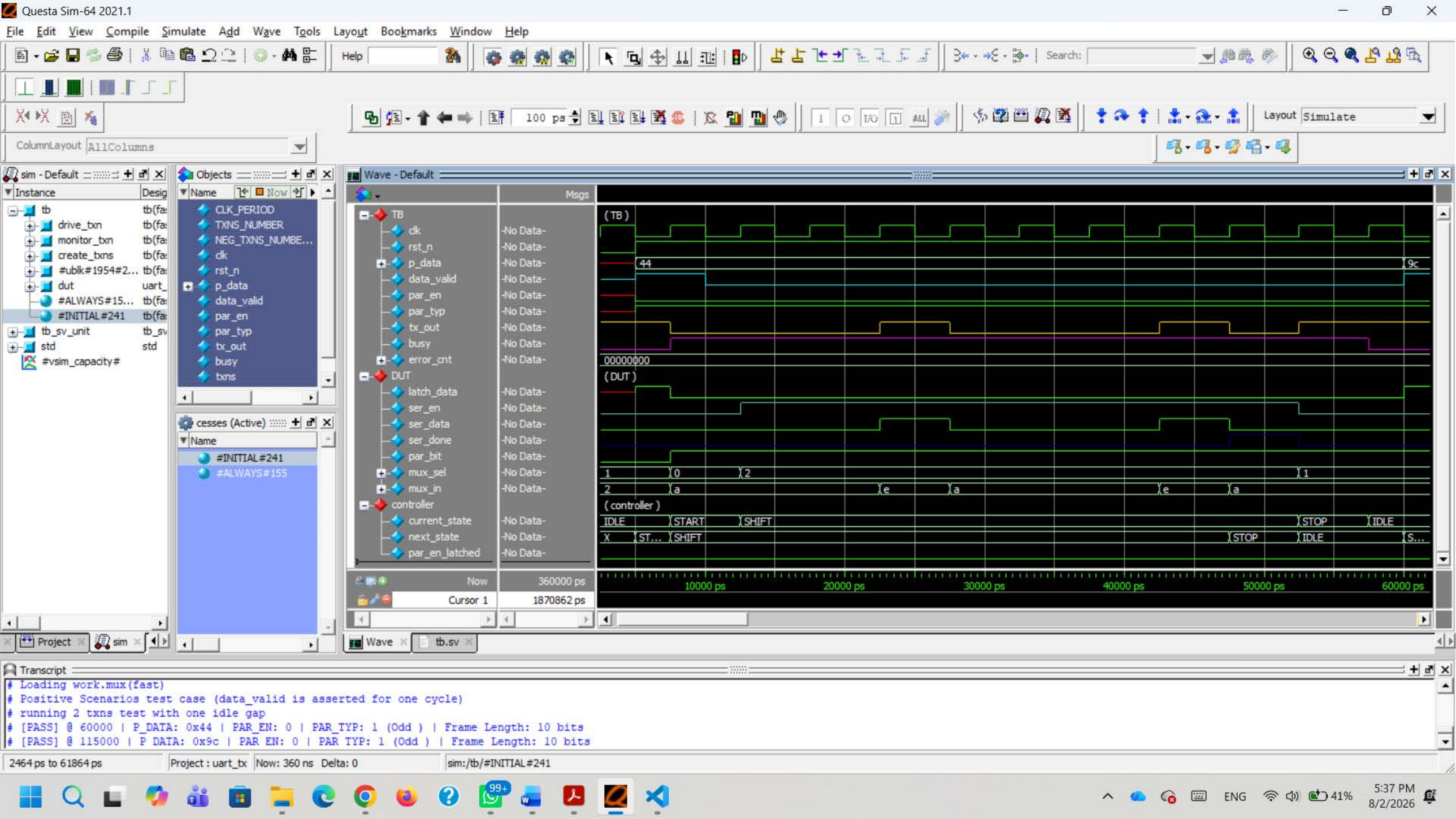
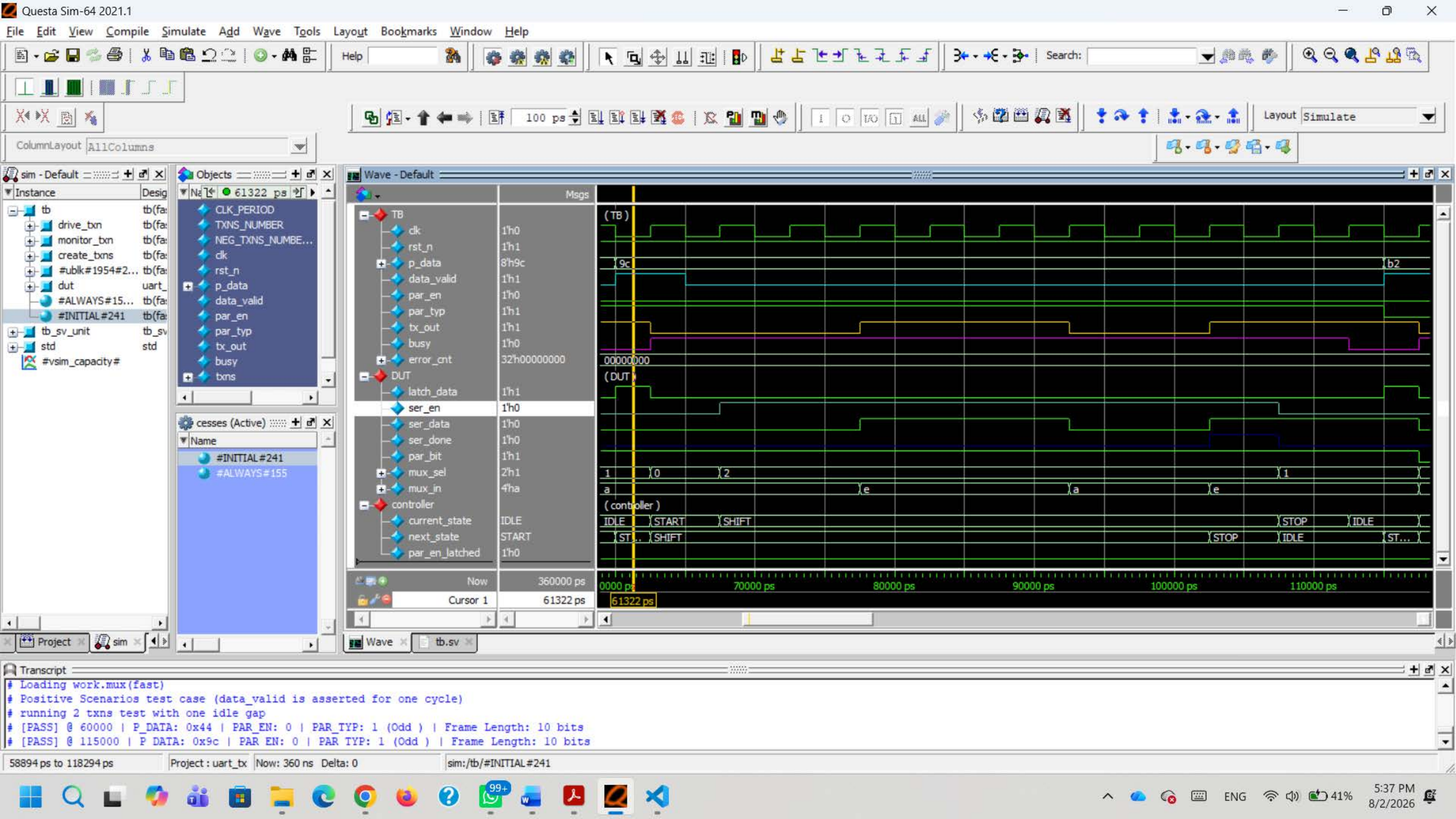
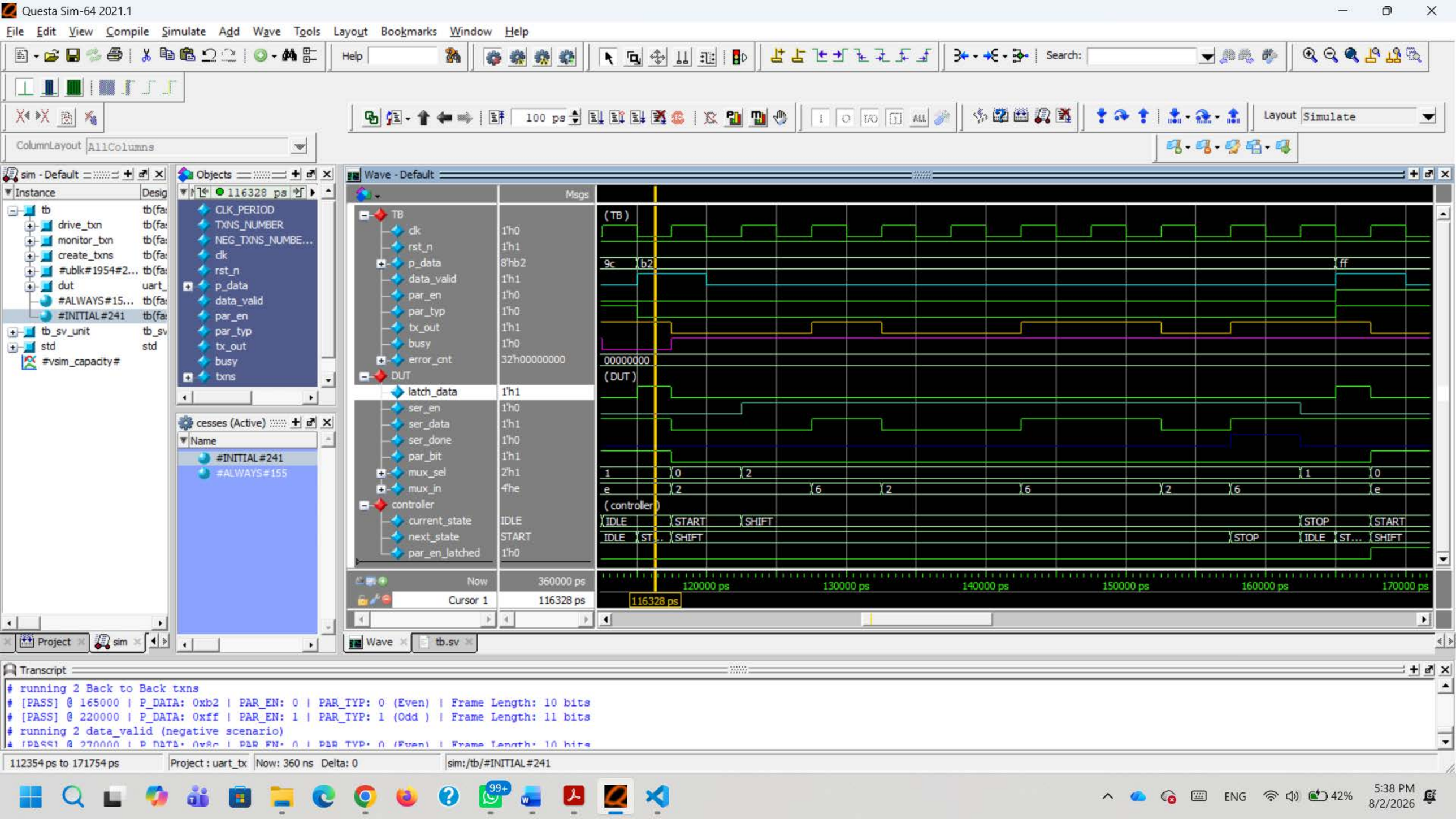
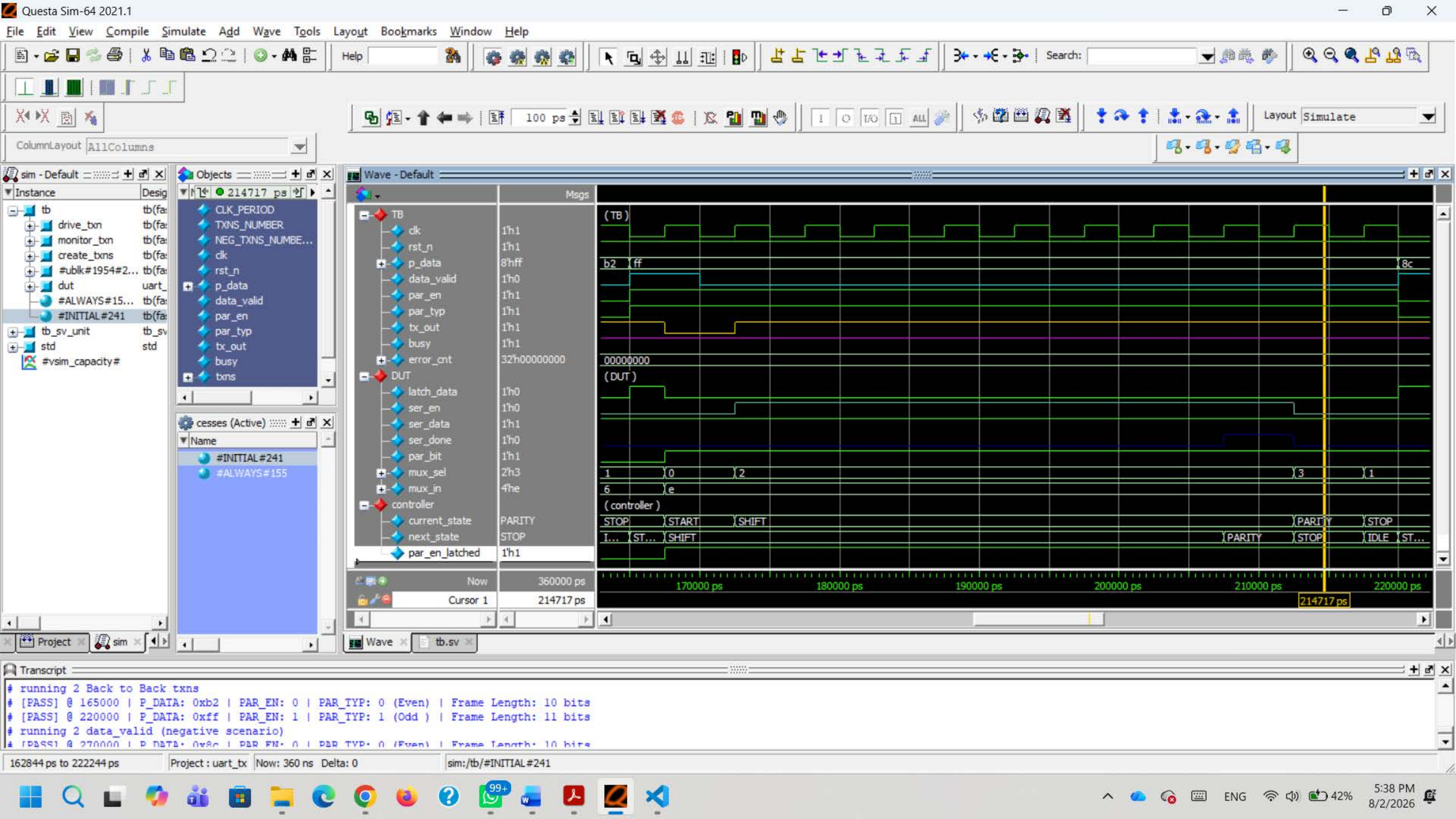
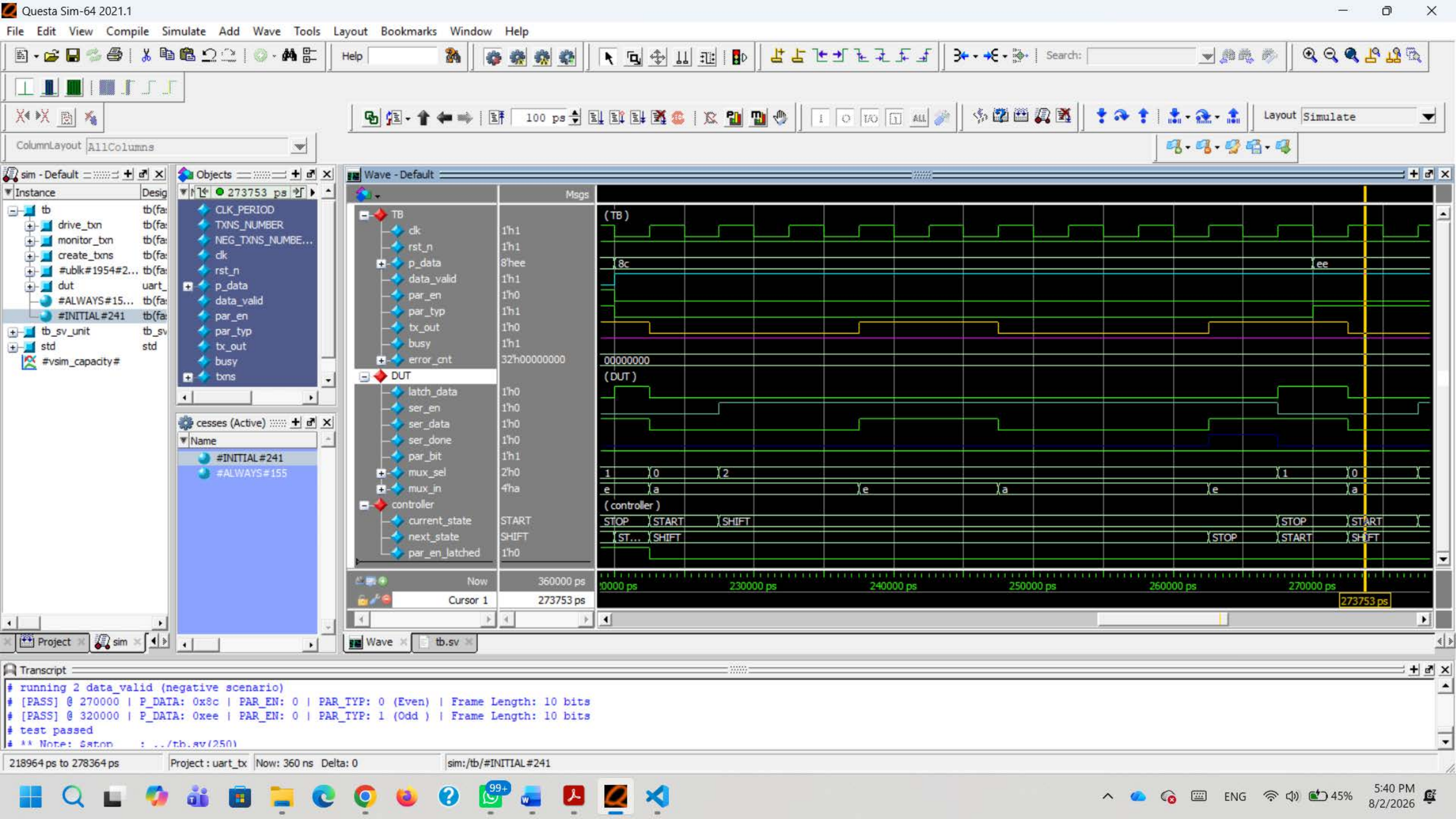
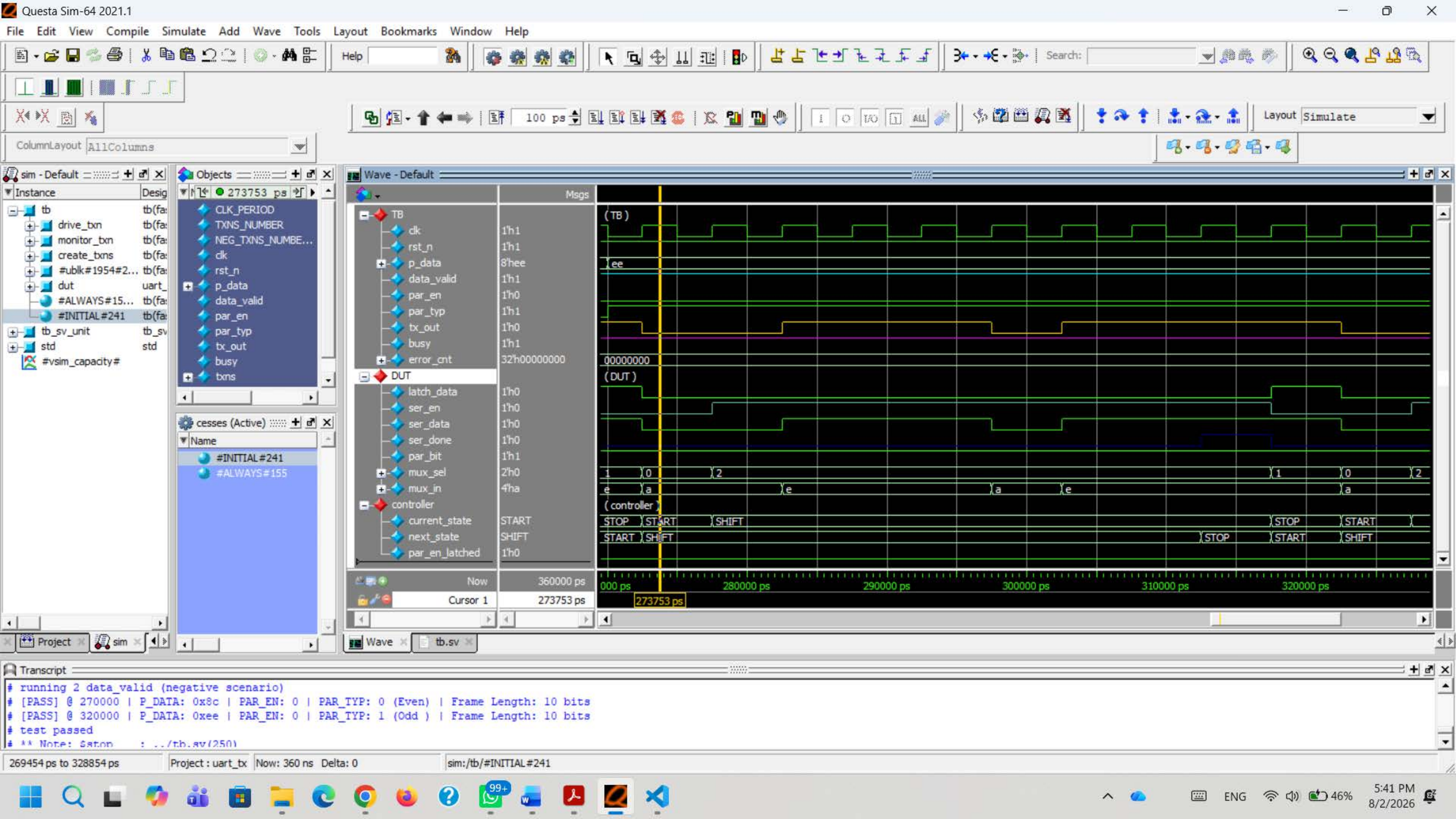

Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
Name
Now
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
error_cnt
DUT
latch_data
ser_en
ser_data
ser_done
par_bit
mux_sel
mux_in
controller
current_state
next_state
par_en_latched
-No Data-
(TB)
44
9c
00000000
(DUT)
(controller)
IDLE
START
SHIFT
STOP
Now
360000 ps
Cursor 1
1870862 ps
10000 ps
20000 ps
30000 ps
40000 ps
50000 ps
60000 ps
Project
sim
Wave
tb.sv
Transcript
Loading work.mux(fast)
Positive Scenarios test case (data_valid is asserted for one cycle)
running 2 txns test with one idle gap
[PASS] @ 60000 | P_DATA: 0x44 | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
[PASS] @ 115000 | P_DATA: 0x9c | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
2464 ps to 61864 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
41%
5:37 PM
8/2/2026

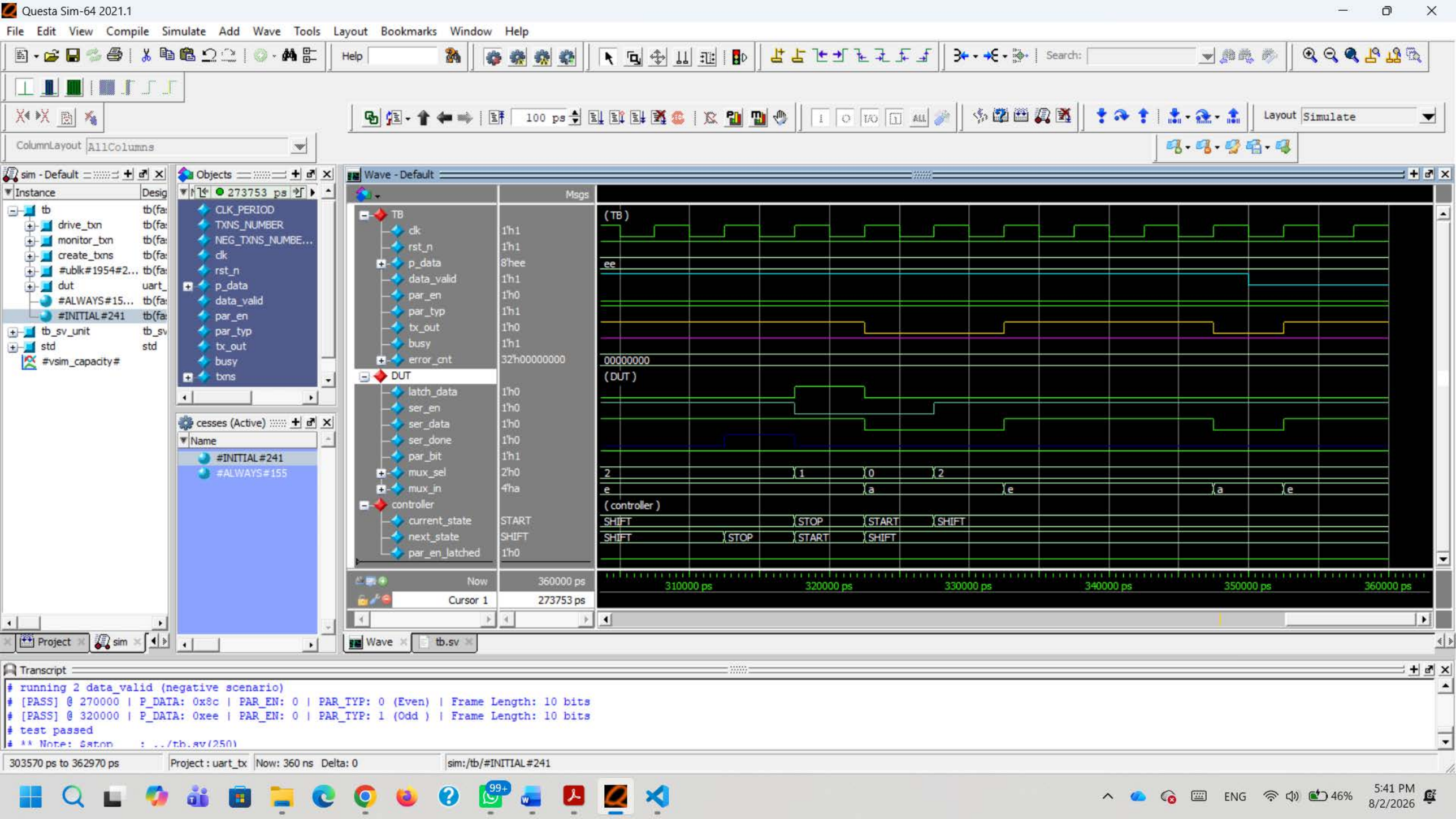

Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
61322 ps
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
Name
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk 1'h0
rst_n 1'h1
p_data 8'h9c
data_valid 1'h1
par_en 1'h0
par_typ 1'h1
tx_out 1'h1
busy 1'h0
error_cnt 32'h00000000
DUT
latch_data 1'h1
ser_en 1'h0
ser_data 1'h0
ser_done 1'h0
par_bit 1'h1
mux_sel 2'h1
mux_in 4'ha
controller
current_state IDLE
next_state START
par_en_latched 1'h0
Now 360000 ps
Cursor 1 61322 ps
0000 ps 70000 ps 80000 ps 90000 ps 100000 ps 110000 ps
61322 ps
Project sim
Wave tb.sv
Transcript
Loading work.mux(fast)
Positive Scenarios test case (data_valid is asserted for one cycle)
running 2 txns test with one idle gap
[PASS] @ 60000 | P_DATA: 0x44 | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
[PASS] @ 115000 | P_DATA: 0x9c | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
58894 ps to 118294 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
41%
5:37 PM
8/2/2026


Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
116328 ps
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
Name
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk 1'h0
rst_n 1'h1
p_data 8'hb2
data_valid 1'h1
par_en 1'h0
par_typ 1'h0
tx_out 1'h1
busy 1'h0
error_cnt 32'h00000000
DUT
latch_data 1'h1
ser_en 1'h0
ser_data 1'h1
ser_done 1'h0
par_bit 1'h1
mux_sel 2'h1
mux_in 4'he
controller
current_state IDLE
next_state START
par_en_latched 1'h0
Now 360000 ps
Cursor 1 116328 ps
(TB)
9c b2 ff
00000000
(DUT)
(controller)
IDLE START SHIFT STOP START
IDLE ST.. SHIFT STOP IDLE ST... SHIFT
120000 ps 130000 ps 140000 ps 150000 ps 160000 ps 170000 ps
116328 ps
Project sim
Wave tb.sv
Transcript
running 2 Back to Back txns
[PASS] @ 165000 | P_DATA: 0xb2 | PAR_EN: 0 | PAR_TYP: 0 (Even) | Frame Length: 10 bits
[PASS] @ 220000 | P_DATA: 0xff | PAR_EN: 1 | PAR_TYP: 1 (Odd) | Frame Length: 11 bits
running 2 data_valid (negative scenario)
112354 ps to 171754 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
42%
5:38 PM
8/2/2026


Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
214717 ps
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
Name
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk 1'h1
rst_n 1'h1
p_data 8'hff
data_valid 1'h0
par_en 1'h1
par_typ 1'h1
tx_out 1'h1
busy 1'h1
error_cnt 32'h00000000
DUT
latch_data 1'h0
ser_en 1'h0
ser_data 1'h1
ser_done 1'h0
par_bit 1'h1
mux_sel 2'h3
mux_in 4'he
controller
current_state PARITY
next_state STOP
par_en_latched 1'h1
Now 360000 ps
Cursor 1 214717 ps
(TB)
b2 ff 8c
00000000
(DUT)
1 0 2 3 1
6 e
(controller)
STOP START SHIFT PARITY STOP
I... ST... SHIFT PARITY STOP IDLE ST...
170000 ps 180000 ps 190000 ps 200000 ps 210000 ps 220000 ps
214717 ps
Project sim
Wave tb.sv
Transcript
running 2 Back to Back txns
[PASS] @ 165000 | P_DATA: 0xb2 | PAR_EN: 0 | PAR_TYP: 0 (Even) | Frame Length: 10 bits
[PASS] @ 220000 | P_DATA: 0xff | PAR_EN: 1 | PAR_TYP: 1 (Odd) | Frame Length: 11 bits
running 2 data_valid (negative scenario)
162844 ps to 222244 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
42%
5:38 PM
8/2/2026


Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
273753 ps
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
Name
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk 1'h1
rst_n 1'h1
p_data 8'hee
data_valid 1'h1
par_en 1'h0
par_typ 1'h1
tx_out 1'h0
busy 1'h1
error_cnt 32'h00000000
DUT
latch_data 1'h0
ser_en 1'h0
ser_data 1'h0
ser_done 1'h0
par_bit 1'h1
mux_sel 2'h0
mux_in 4'ha
controller
current_state START
next_state SHIFT
par_en_latched 1'h0
Now 360000 ps
Cursor 1 273753 ps
(TB)
8c
ee
00000000
(DUT)
(controller)
STOP START SHIFT
230000 ps 240000 ps 250000 ps 260000 ps 270000 ps
273753 ps
Wave
tb.sv
Transcript
running 2 data_valid (negative scenario)
[PASS] @ 270000 | P_DATA: 0x8c | PAR_EN: 0 | PAR_TYP: 0 (Even) | Frame Length: 10 bits
[PASS] @ 320000 | P_DATA: 0xee | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
test passed
218964 ps to 278364 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
45%
5:40 PM
8/2/2026


Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
273753 ps
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
Name
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk 1'h1
rst_n 1'h1
p_data 8'hee
data_valid 1'h1
par_en 1'h0
par_typ 1'h1
tx_out 1'h0
busy 1'h1
error_cnt 32'h00000000
DUT
latch_data 1'h0
ser_en 1'h0
ser_data 1'h0
ser_done 1'h0
par_bit 1'h1
mux_sel 2'h0
mux_in 4'ha
controller
current_state START
next_state SHIFT
par_en_latched 1'h0
Now 360000 ps
Cursor 1 273753 ps
000 ps
280000 ps
290000 ps
300000 ps
310000 ps
320000 ps
273753 ps
(TB)
ee
00000000
(DUT)
(controller)
STOP START SHIFT
START SHIFT STOP START SHIFT
Project
sim
Wave
tb.sv
Transcript
running 2 data_valid (negative scenario)
[PASS] @ 270000 | P_DATA: 0x8c | PAR_EN: 0 | PAR_TYP: 0 (Even) | Frame Length: 10 bits
[PASS] @ 320000 | P_DATA: 0xee | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
test passed
269454 ps to 328854 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
46%
5:41 PM
8/2/2026


Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
drive_txn
monitor_txn
create_txns
#ublk#1954#2...
dut
#ALWAYS#15...
#INITIAL#241
tb_sv_unit
std
#vsim_capacity#
Objects
273753 ps
CLK_PERIOD
TXNS_NUMBER
NEG_TXNS_NUMBE...
clk
rst_n
p_data
data_valid
par_en
par_typ
tx_out
busy
txns
cesses (Active)
Name
#INITIAL#241
#ALWAYS#155
Wave - Default
Msgs
TB
clk 1'h1
rst_n 1'h1
p_data 8'hee
data_valid 1'h1
par_en 1'h0
par_typ 1'h1
tx_out 1'h0
busy 1'h1
error_cnt 32'h00000000
DUT
latch_data 1'h0
ser_en 1'h0
ser_data 1'h0
ser_done 1'h0
par_bit 1'h1
mux_sel 2'h0
mux_in 4'ha
controller
current_state START
next_state SHIFT
par_en_latched 1'h0
Now 360000 ps
Cursor 1 273753 ps
(TB)
ee
00000000
(DUT)
2 1 0 2
e a e a e
(controller)
SHIFT STOP START SHIFT
SHIFT STOP START SHIFT
310000 ps 320000 ps 330000 ps 340000 ps 350000 ps 360000 ps
Project sim
Wave tb.sv
Transcript
running 2 data_valid (negative scenario)
[PASS] @ 270000 | P_DATA: 0x8c | PAR_EN: 0 | PAR_TYP: 0 (Even) | Frame Length: 10 bits
[PASS] @ 320000 | P_DATA: 0xee | PAR_EN: 0 | PAR_TYP: 1 (Odd) | Frame Length: 10 bits
test passed
303570 ps to 362970 ps
Project : uart_tx
Now: 360 ns Delta: 0
sim:/tb/#INITIAL#241
ENG
46%
5:41 PM
8/2/2026